DSC 140B - Quiz 07

February 26, 2026

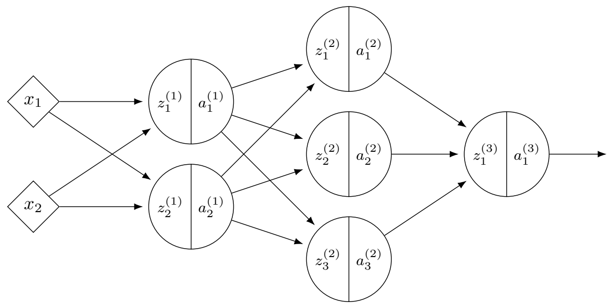
Name: PID:

About the quizzes:

- Quizzes in DSC 140B are *optional* and graded pass/fail.
- A score of 70% or higher earns a "pass" and 1.5 credits toward your final grade.
- If you don't pass, no credits are earned, but it doesn't hurt your grade.
- You have 30 minutes to complete the quiz.
- At least one of the questions below will be on an exam (probably with slight changes, such as different numbers).
- Unfortunately, we can't answer clarifying questions during the quiz. If you think a question has a bug or is unclear, please let us known in a private post on Campuswire after the quiz, and we'll take it into account when grading.

Problem 1. (4 points)

Consider a neural network $H(\vec{x})$ shown below:



Assume that all hidden nodes use **ReLU activation** functions, that the output node uses a linear activation, and that there are no biases.

Let the weights of the second layer of the network be: $W^{(2)} = \begin{pmatrix} 3 & -1 & 2 \\ 1 & 4 & -2 \end{pmatrix}$.

In addition, suppose it is known that, when the network is evaluated at $\vec{x}$ with weight vector $\vec{w}$:

$$z_1^{(1)} = 4, \qquad z_2^{(1)} = -1$$

$$\partial H/\partial z_1^{(2)} = 2, \qquad \partial H/\partial z_2^{(2)} = -1, \qquad \partial H/\partial z_3^{(2)} = 3$$

a) What is $\partial H/\partial W_{11}^{(2)}$, as a number?

b) What is $\partial H/\partial W_{21}^{(2)}$, as a number?

c) What is $\partial H/\partial a_1^{(1)}$, as a number?

d) What is $\partial H/\partial z_2^{(1)}$, as a number?

Problem 2.

Consider a deep neural network $H(\vec{x})$ whose output layer uses a **sigmoid** activation function. For a particular input $\vec{x}$, the network's output is $H(\vec{x}) \approx 0$ (that is, it is approximately zero).

True or False: for this input, $\partial H / \partial W_{ij}^{(\ell)} \approx 0$ for all i, j, ℓ.

- ◯ True
- ◯ False

Problem 3. (2 points)

Recall that the gradient of the square loss at a single data point $(\vec{x}, y)$ is:

$$\nabla_{\vec{w}} L = 2(\text{Aug}(\vec{x}) \cdot \vec{w} - y)\,\text{Aug}(\vec{x})$$

Suppose you are running **stochastic** gradient descent to minimize the empirical risk with respect to the square loss in order to train a function $H(x) = w_0 + w_1 x$ on the following data set:

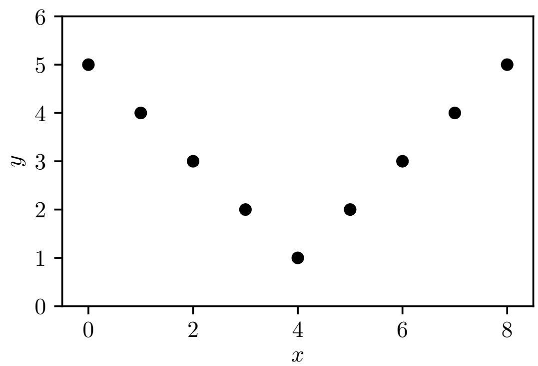
x	y
2	3
1	-1
3	5
4	7
5	2

Suppose that the initial weight vector is $\vec{w} = (0, 0)^T$, the learning rate is $\eta = 1/2$, and the mini-batch size is 2. If the first mini-batch consists of the 2nd and 3rd data points, what will be the weight vector after one iteration of stochastic gradient descent?

Hint: the first entry should be 2.

Problem 4.

Suppose you want to train a neural network with a **single hidden layer** to predict y from x on the following one-dimensional regression data set:



True or False: if the hidden layer has sufficiently many nodes and uses **linear** activations (with a linear output node), the network can achieve zero empirical risk on this data set.

◯ True

◯ False

Problem 5.

Suppose you are training a neural network with two hidden layers of 64 nodes each, using ReLU activations.

True or False: it is typical to initialize all of the weights of this network to zero before training.

◯ True

◯ False

Problem 6.

Suppose you are training a deep neural network with 5 hidden layers, each using ReLU activations, to perform regression. You use stochastic gradient descent with a fixed learning rate to minimize the empirical risk with respect to the squared loss.

True or False: if trained for sufficiently many epochs, SGD is guaranteed to find a global minimum of the empirical risk.

◯ True

◯ False